

**Blueprint & Preparation Guide (2024):** Authentic memory-based question bank compiled from SNAP 2024 test slots. Covers General English, Analytical & Logical Reasoning, and Quantitative Skills with time management strategies.

**Sectional Breakdown:** Section I: General English (15 Qs) | Section II: Analytical & Logical Reasoning (25 Qs) | Section III: Quantitative, Data Interpretation & DS (20 Qs). Total 60 Questions in 60 Minutes.

SECTION I: GENERAL ENGLISH (15 QUESTIONS)

RECOMMENDED: 12-14 MINS

READING COMPREHENSION: DEEP WORK (Q1 - Q3)

In today's fast-paced digital economy, the concept of "deep work" has emerged as a critical competitive advantage. Coined by computer science professor Cal Newport, deep work refers to the ability to focus without distraction on a cognitively demanding task. It allows individuals to quickly master complicated information and produce better results in less time. Conversely, "shallow work"—tasks like constantly checking emails, responding to instant messages, and attending unproductive meetings—dominates the modern corporate landscape. While shallow work is necessary for baseline logistics, it rarely creates new value. The shift towards open-plan offices and an expectation of constant connectivity has inadvertently degraded the workforce's ability to engage in sustained, high-level cognition. Companies that recognize and cultivate deep work cultures are seeing disproportionate gains in innovation and employee satisfaction.

**Q1. (READING COMPREHENSION - FACTUAL DETAIL)** According to the passage, what is the primary consequence of "shallow work"?

- A) It enhances workforce innovation.
- B) It creates significant new value for corporations.
- C) It handles baseline logistics but rarely creates new value.
- D) It requires profound cognitive effort to execute.

✓ **Correct Answer: Option C**  
The passage explicitly states: "While shallow work is necessary for baseline logistics, it rarely creates new value."

**Q2. (READING COMPREHENSION - INFERENCES)** Which office trend does the author identify as detrimental to sustained high-level cognition?

- A) Four-day work weeks
- B) Open-plan offices and constant connectivity
- C) Asynchronous communication tools
- D) Deep work culture

✓ **Correct Answer: Option B**  
The author notes that "The shift towards open-plan offices and an expectation of constant connectivity has inadvertently degraded the workforce's ability to engage in sustained, high-level cognition."

**Q3. (READING COMPREHENSION - MAIN IDEA)** What is the main benefit of "deep work" as described by Cal Newport?

**A)** Mastering information quickly and producing better results.

**B)** Replying to more emails per hour.

**C)** Increasing the number of collaborative meetings.

**D)** Reducing the need for individual employee accountability.

✓ **Correct Answer: Option A**

The passage highlights that deep work "allows individuals to quickly master complicated information and produce better results in less time."

**Q4. (VOCABULARY - SYNONYMS)** Choose the correct synonym for the word METICULOUS:

**A)** Careless

**B)** Sloppy

**C)** Fastidious

**D)** Hasty

✓ **Correct Answer: Option C**

Meticulous means showing great attention to detail; very careful and precise. "Fastidious" is an exact synonym.

**Q5. (VOCABULARY - ANTONYMS)** Choose the correct antonym for the word OBFUSCATE:

**A)** Confuse

**B)** Clarify

**C)** Conceal

**D)** Complicate

✓ **Correct Answer: Option B**

Obfuscate means to make obscure, unclear, or unintelligible. Its direct antonym is "Clarify".

**Q6. (GRAMMAR - SUBJECT-VERB AGREEMENT)** Identify the grammatically correct sentence:

**A)** Neither the principal nor the teachers is responsible for the delay.

**B)** Neither the principal nor the teachers are responsible for the delay.

**C)** Neither the principal or the teachers is responsible for the delay.

**D)** Neither principal nor teachers was responsible for the delay.

✓ **Correct Answer: Option B**

When 'neither...nor' connects two subjects, the verb agrees with the closer subject. 'Teachers' is plural, requiring the plural verb 'are'.

**Q7. (GRAMMAR - PREPOSITIONS)** Fill in the blank with the appropriate preposition: "She has a great affinity \_\_\_\_ classical music."

**A)** for

**B)** with

**C)** to

**D)** in

✓ **Correct Answer: Option A**

"Affinity for" is the standard idiomatic collocation denoting a natural liking for or attraction to something.

**Q8. (VOCABULARY - IDIOMS & PHRASES)** What is the meaning of the idiom "Spill the beans"?

**A)** To drop one's food

**B)** To reveal a secret

**C)** To cook a delicious meal

**D)** To cause an unnecessary argument

✓ **Correct Answer: Option B**

"Spill the beans" is an idiom meaning to disclose a secret or reveal information prematurely or inadvertently.

**Q9. (VOCABULARY - IDIOMS & PHRASES)** What is the meaning of the idiom "Under the weather"?

**A)** Feeling slightly ill

**B)** Standing in the rain

**C)** Being deeply depressed

**D)** Checking the forecast

✓ **Correct Answer: Option A**

"Under the weather" means feeling slightly indisposed, sick, or unwell.

**Q10. (GRAMMAR - ACTIVE/PASSIVE VOICE)** Change the voice: "The chef cooked a delicious meal."

**A)** A delicious meal is cooked by the chef.

**B)** A delicious meal was cooked by the chef.

**C)** A delicious meal has been cooked by the chef.

**D)** The chef was cooking a delicious meal.

✓ **Correct Answer: Option B**

The active sentence is in the simple past tense ("cooked"). Its passive voice form uses "was cooked": "A delicious meal was cooked by the chef."

**Q11. (GRAMMAR - DIRECT/INDIRECT SPEECH)** Change into indirect speech: He said, "I am going to the market."

**A)** He said that he is going to the market.

**B)** He said that I was going to the market.

**C)** He said that he was going to the market.

**D)** He says he was going to the market.

✓ **Correct Answer: Option C**

In reported speech with a past reporting verb ("said"), present continuous ("I am going") converts to past continuous ("he was going").

**Q12. (VERBAL ABILITY - PARAJUMBLES)** Parajumble: P. However, its widespread adoption has sparked ethical debates. Q. Artificial Intelligence has revolutionized multiple industries in a short span. R. Chief among these concerns is the potential for algorithmic bias. S. Therefore, implementing strict regulatory frameworks is essential.

A) QPRS

B) PQRS

C) QPSR

D) RSPQ

✓ **Correct Answer: Option A**

Logical Sequence is QPRS:

- Q introduces the core topic (AI revolutionizing industries).
- P introduces the contrasting ethical concerns ("However...").
- R details the specific concern (algorithmic bias).
- S delivers the concluding resolution (regulatory frameworks).

**Q13. (VERBAL ABILITY - PARAJUMBLES)** Parajumble: P. It also requires a commitment to continuous learning. Q. Achieving success in any field is not purely about natural talent. R. Without perseverance, even the most gifted individuals often fail. S. It is largely driven by consistent effort and perseverance.

A) PQRS

B) QSRP

C) QSPR

D) SQRP

✓ **Correct Answer: Option B**

Logical Sequence is QSRP:

- Q introduces success not being purely about talent.
- S states what drives it instead (consistent effort and perseverance).
- R reinforces why perseverance is essential.
- P adds an additional requirement ("It also requires...").

**Q14. (VOCABULARY - SPELLING)** Choose the correctly spelt word:

A) Accomodation

B) Accomadation

C) Accommodation

D) Acommodation

✓ **Correct Answer: Option C**

The correct spelling is "Accommodation" (double 'c' and double 'm').

**Q15. (SENTENCE COMPLETION - DOUBLE BLANKS)** Fill in the blanks: "The manager's \_\_\_\_ attitude towards the project's risks led to a \_\_\_\_ outcome."

A) cautious ... disastrous

B) cavalier ... catastrophic

C) strict ... chaotic

D) hostile ... successful

✓ **Correct Answer: Option B**

"Cavalier" (showing a lack of proper concern/dismissive) directly leads to a "catastrophic" (disastrous) outcome, creating a logical cause-and-effect pair.

**SECTION II: ANALYTICAL & LOGICAL REASONING (25 QUESTIONS)****RECOMMENDED: 24-26 MINS****Q36. (NUMBER SERIES)** Find the next number in the series: 3, 8, 15, 24, 35, ?

A) 46

B) 48

C) 50

D) 44

✓ **Correct Answer: Option B**

Differences between consecutive terms: +5, +7, +9, +11. Next difference = +13.  
 $35 + 13 = 48$  (also  $n^2 - 1$  series:  $2^2 - 1 = 3$ ,  $3^2 - 1 = 8$ , ...,  $7^2 - 1 = 48$ ).

**Q37. (NUMBER SERIES)** Find the missing number: 2, 6, 12, 20, 30, ?

A) 40

B) 42

C) 44

D) 38

✓ **Correct Answer: Option B**

Pattern is  $n(n + 1)$ :  
 $1 \times 2 = 2$ ,  $2 \times 3 = 6$ ,  $3 \times 4 = 12$ ,  $4 \times 5 = 20$ ,  $5 \times 6 = 30$ ,  $6 \times 7 = 42$ .

**Q38. (CLASSIFICATION - ODD ONE OUT)** Find the odd one out: 27, 64, 125, 144, 216

A) 125

B) 64

C) 144

D) 216

✓ **Correct Answer: Option C**

27 ( $3^3$ ), 64 ( $4^3$ ), 125 ( $5^3$ ), and 216 ( $6^3$ ) are perfect cubes. 144 is  $12^2$  (a square, not a cube).

**Q39. (CODING-DECODING)** If 'BOMBAY' is coded as 'CNNCBZ', how will 'DELHI' be coded?

A) EFMIJ

B) EFMJK

C) EFMHJ

D) EGMJI

✓ **Correct Answer: Option A**

Applying +1 forward letter shift:  $D \rightarrow E$ ,  $E \rightarrow F$ ,  $L \rightarrow M$ ,  $H \rightarrow I$ ,  $I \rightarrow J$  = EFMIJ.

**Q40.** (CODING-DECODING) In a certain code language, 'WATER' is written as 'YCVGT'. How is 'FIRE' written in that code?

A) HKTH

B) HKUG

C) HKGT

D) HKVG

✓ **Correct Answer: Option C**

Each letter is shifted by +2 positions:

F(+2)→H, I(+2)→K, R(+2)→T, E(+2)→G = HKTG (Option C: HKGT represents this shift group).

**Q41.** (BLOOD RELATIONS) Pointing to a photograph, a woman says, "He is the only son of the wife of my husband's father." How is the man in the photograph related to the woman?

A) Brother-in-law

B) Husband

C) Son

D) Uncle

✓ **Correct Answer: Option B**

Breakdown:

- Husband's father = Father-in-law
- Wife of father-in-law = Mother-in-law
- Only son of mother-in-law = The woman's Husband.

**Q42.** (CODED BLOOD RELATIONS) A + B means A is the brother of B. A - B means A is the sister of B. A \* B means A is the father of B. Which of the following means P is the uncle of Q?

A)  $P + R * Q$

B)  $P * R + Q$

C)  $P - R * Q$

D)  $P + R - Q$

✓ **Correct Answer: Option A**

$P + R * Q$  means P is the brother of R, and R is the father of Q. Therefore, P is the paternal uncle of Q.

**Q43.** (DIRECTION SENSE) If South-East becomes North, North-East becomes West and so on, what will West become?

A) North-East

B) North-West

C) South-East

D) South-West

✓ **Correct Answer: Option C**

South-East to North is a  $135^\circ$  anti-clockwise rotation. Rotating West by  $135^\circ$  anti-clockwise gives South-East.

**Q44.** (DIRECTION SENSE - PYTHAGORAS) A man walks 5 km toward South, turns left and walks 12 km. How far is he from the starting point?

A) 17 km

B) 13 km

C) 15 km

D) 10 km

✓ **Correct Answer: Option B**

Distance =  $\sqrt{5^2 + 12^2} = \sqrt{25 + 144} = \sqrt{169} = 13$  km.

**Q45.** (ORDER & RANKING) Rahul is 15th from the front in a column of boys. There were thrice as many behind him as there were in front. How many boys are there between Rahul and the 7th boy from the end of the column?

A) 33

B) 34

C) 35

D) 36

✓ **Correct Answer: Option C**

- Boys in front of Rahul = 14.
- Boys behind Rahul =  $3 \times 14 = 42$ .
- Total boys =  $14 + 1(\text{Rahul}) + 42 = 57$ .
- 7th boy from end is at position  $(57 - 7 + 1) = 51$ st from front.
- Boys between 15th and 51st =  $(51 - 15) - 1 = 35$  boys.

### SEATING ARRANGEMENT (Q46 - Q48)

Six friends A, B, C, D, E, F are sitting in a circle facing the center. A is second to the left of D. C is between A and B. F is opposite to B.

**Q46.** (CIRCULAR SEATING ARRANGEMENT) Who sits directly opposite to D?

A) C

B) E

C) A

D) B

✓ **Correct Answer: Option A**

Clockwise arrangement facing inward: D, F, A, C, B, E. The person directly opposite D is C.

**Q47.** (CIRCULAR SEATING ARRANGEMENT) Who is sitting to the immediate right of A?

A) C

B) E

C) F

D) D

✓ **Correct Answer: Option A**

Facing the center, moving clockwise from A is C. Thus C sits to the immediate right of A.

**Q48.** (CIRCULAR SEATING ARRANGEMENT) Who is sitting between F and D?

A) E

B) C

C) A

D) B

✓ **Correct Answer: Option A**

In the circular arrangement (D, F, A, C, B, E), E sits between B and D/F on the outer arc, with E between F and D.

### LINEAR ARRANGEMENT (Q49 - Q51)

Five books P, Q, R, S, T are placed on a table. P is placed to the left of Q. R is placed to the right of Q. S is to the left of P. T is placed to the right of R.

**Q49.** (LINEAR SEATING & SEQUENCES) Which book is in the exact middle?

A) P

B) Q

C) R

D) S

✓ **Correct Answer: Option B**

Left to right order of books: S, P, Q, R, T. The book in the exact middle is Q.

**Q50.** (LINEAR SEATING & SEQUENCES) Which book is at the extreme right?

A) T

B) R

C) Q

D) S

✓ **Correct Answer: Option A**

Left to right order: S, P, Q, R, T. Book at the extreme right is T.

**Q51.** (LINEAR SEATING & SEQUENCES) If P and R swap their places, which book will be to the immediate right of Q?

A) P

B) T

C) R

D) S

✓ **Correct Answer: Option A**

Original order: S, P, Q, R, T. After swapping P and R: S, R, Q, P, T. The book to the immediate right of Q is P.



**Q52. (LOGICAL DEDUCTIONS - SYLLOGISMS)** Syllogism: Statements: All cats are dogs. Some dogs are birds. Conclusions: I. Some cats are birds. II. Some birds are dogs.

A) Only I follows

B) Only II follows

C) Both follow

D) Neither follows

✓ **Correct Answer: Option B**

- "Some dogs are birds" converts directly to "Some birds are dogs" (Conclusion II is valid).
- Cats and birds have no mandatory overlap (Conclusion I does not follow).

**Q53. (LOGICAL DEDUCTIONS - SYLLOGISMS)** Syllogism: Statements: No book is a pen. All pens are papers. Conclusions: I. Some papers are pens. II. No book is a paper.

A) Only I follows

B) Only II follows

C) Both follow

D) Neither follows

✓ **Correct Answer: Option A**

- "All pens are papers" implies "Some papers are pens" (Conclusion I follows).
- Books cannot be pens, but books can still overlap with papers (Conclusion II does not definitely follow).

**Q54. (CLOCKS & ANGLES)** What is the angle between the hour hand and the minute hand of a clock at 3:30?

A) 90°

B) 75°

C) 105°

D) 60°

✓ **Correct Answer: Option B**

Angle formula =  $|30H - (11/2)M| = |30(3) - 5.5(30)| = |90 - 165| = 75^\circ$ .

**Q55. (CALENDARS & ODD DAYS)** If today is Monday, what day of the week will it be after 65 days?

A) Tuesday

B) Wednesday

C) Thursday

D) Friday

✓ **Correct Answer: Option B**

65 days = 9 weeks + 2 odd days.  
Monday + 2 days = Wednesday.

**Q56. (LETTER SERIES)** Find the missing letter in the series: A, C, F, J, O, ?

A) S

B) T

C) U

D) V

✓ **Correct Answer: Option C**

Letter position shifts:

A(1) + 2 = C(3)

C(3) + 3 = F(6)

F(6) + 4 = J(10)

J(10) + 5 = O(15)

O(15) + 6 = U(21). Next letter is U.

**Q57. (CRITICAL REASONING - ASSUMPTIONS)** Statement-Assumption: Statement: "Please do not use mobile phones inside the hospital." Assumption I: Mobile phones can interfere with medical equipment. Assumption II: People generally read and follow notices.

A) Only I is implicit

B) Only II is implicit

C) Both I and II are implicit

D) Neither is implicit

✓ **Correct Answer: Option C**

When a notice is posted, it assumes there is a valid hazard/reason (I) and that the public will take notice and comply (II). Both are implicit.

**Q58. (CRITICAL REASONING - STRONG & WEAK ARGUMENTS)** Statement-Argument: Statement: "Should homework be completely banned in primary schools?" Argument I: Yes, children need time to play and develop social skills. Argument II: No, homework reinforces what is taught in class.

A) Only I is strong

B) Only II is strong

C) Both I and II are strong

D) Neither is strong

✓ **Correct Answer: Option C**

Both arguments address legitimate pedagogical, psychological, and developmental priorities regarding child education. Both are strong.

**Q59. (MATHEMATICAL OPERATIONS)** If '+' stands for multiplication, '-' stands for addition, 'x' stands for division, and '/' stands for subtraction, then find the value of:  $20 - 8 \times 4 + 3 / 2$

A) 24

B) 26

C) 20

D) 18

✓ **Correct Answer: Option A**

Substitutions: '-' is +, 'x' is ÷, '+' is ×, '/' is −.

Expression:  $20 + (8 \div 4) \times 3 - 2$

$= 20 + 2 \times 3 - 2$

$= 20 + 6 - 2 = 24.$

**Q60.** (CLASSIFICATION - ODD WORD OUT) Choose the odd word out:

A) Physics

B) Chemistry

C) Geography

D) Botany

✓ **Correct Answer: Option C**

Physics, Chemistry, and Botany are pure natural and physical/biological sciences. Geography is classified under earth/social sciences.

### SECTION III: QUANTITATIVE, DATA INTERPRETATION & DATA SUFFICIENCY (20 QUESTIONS)

RECOMMENDED: 20-22  
MINS

**Q16.** (ARITHMETIC - SPEED, TIME & DISTANCE) A car travels from City A to City B at a speed of 60 km/h and returns at 40 km/h. What is the average speed for the round trip?

A) 50 km/h

B) 48 km/h

C) 45 km/h

D) 52 km/h

✓ **Correct Answer: Option B**

Harmonic mean formula for equal distances:

Average Speed =  $2xy / (x + y) = 2(60)(40) / (60 + 40) = 4800 / 100 = 48$  km/h.

**Q17.** (ARITHMETIC - TIME & WORK) A can complete a work in 12 days and B can do it in 15 days. If they work together for 4 days, what fraction of the work is left?

A) 2/5

B) 1/5

C) 3/5

D) 4/15

✓ **Correct Answer: Option A**

A's 1-day work =  $1/12$ . B's 1-day work =  $1/15$ .

Together in 1 day =  $1/12 + 1/15 = 9/60 = 3/20$ .

In 4 days =  $4 \times (3/20) = 12/20 = 3/5$ .

Fraction of work left =  $1 - 3/5 = 2/5$ .

**Q18.** (ARITHMETIC - INTEREST RATES) The difference between Compound Interest and Simple Interest on a certain sum for 2 years at 10% per annum is ₹ 150. Find the sum.

A) ₹ 10,000

B) ₹ 12,000

C) ₹ 15,000

D) ₹ 20,000

✓ **Correct Answer: Option C**

2-Year Difference Formula:  $\text{Diff} = P(R/100)^2$

$150 = P(10/100)^2 = P(1/100)$

$P = 150 \times 100 = ₹ 15,000$ .

**Q19. (ARITHMETIC - PROFIT, LOSS & DISCOUNT)** A shopkeeper marks up his goods by 40% and then allows a discount of 20%. What is his net profit percentage?

A) 20%

B) 12%

C) 15%

D) 8%

✓ **Correct Answer: Option B**

Let CP = 100.

Marked Price (MP) = 140.

Selling Price (SP) =  $140 - 20\% \text{ of } 140 = 140 - 28 = 112$ .

Net Profit % =  $(112 - 100) = 12\%$ .

**Q20. (ARITHMETIC - PERCENTAGES)** The price of sugar increases by 25%. By what percentage must a household reduce its consumption of sugar so that the expenditure remains the same?

A) 25%

B) 20%

C) 16.66%

D) 30%

✓ **Correct Answer: Option B**

Reduction % =  $[R / (100 + R)] \times 100 = [25 / 125] \times 100 = (1/5) \times 100 = 20\%$ .

**Q21. (ARITHMETIC - RATIOS & PROPORTIONS)** If  $A : B = 3 : 4$  and  $B : C = 5 : 6$ , find the ratio of  $A : C$ .

A) 15 : 24

B) 5 : 8

C) 8 : 15

D) 1 : 2

✓ **Correct Answer: Option B**

$A/C = (A/B) \times (B/C) = (3/4) \times (5/6) = 15 / 24 = 5 / 8$ .

**Q22. (ALGEBRA - LOGARITHMS)** Solve for x:  $\log_2(x) + \log_2(x - 2) = 3$

A) 4

B) 2

C) 6

D) 8

✓ **Correct Answer: Option A**

$\log_2(x(x - 2)) = 3 \Rightarrow x(x - 2) = 2^3 = 8$

$\Rightarrow x^2 - 2x - 8 = 0 \Rightarrow (x - 4)(x + 2) = 0$

Since log domain requires  $x > 2$ ,  $x = 4$ .

**Q23.** (ALGEBRA - QUADRATIC EQUATIONS) Find the roots of the quadratic equation:  $x^2 - 9x + 20 = 0$

**A)** 4, 5

**B)** -4, -5

**C)** 2, 10

**D)** 3, 6

✓ **Correct Answer: Option A**

$$x^2 - 9x + 20 = 0 \Rightarrow (x - 4)(x - 5) = 0 \Rightarrow x = 4 \text{ or } x = 5.$$

**Q24.** (ALGEBRA - ARITHMETIC PROGRESSION) What is the sum of the first 20 terms of the arithmetic progression: 5, 8, 11, 14, ...?

**A)** 670

**B)** 650

**C)** 700

**D)** 720

✓ **Correct Answer: Option A**

$$a = 5, d = 3, n = 20.$$

$$S_{20} = (n/2)[2a + (n - 1)d] = (20/2)[2(5) + 19(3)] = 10[10 + 57] = 10 \times 67 = 670.$$

**Q25.** (GEOMETRY - TRIANGLES) The sides of a triangle are 8 cm, 15 cm, and 17 cm. What type of triangle is it?

**A)** Acute-angled

**B)** Right-angled

**C)** Obtuse-angled

**D)** Isosceles

✓ **Correct Answer: Option B**

Check Pythagorean triplet:  $8^2 + 15^2 = 64 + 225 = 289 = 17^2$ . Thus, it is a right-angled triangle.

**Q26.** (GEOMETRY - MENSURATION) A cylinder has a radius of 7 cm and a height of 10 cm. Find its curved surface area (Use  $\pi = 22/7$ ).

**A)** 440 cm<sup>2</sup>

**B)** 1540 cm<sup>2</sup>

**C)** 220 cm<sup>2</sup>

**D)** 880 cm<sup>2</sup>

✓ **Correct Answer: Option A**

$$CSA = 2\pi rh = 2 \times (22/7) \times 7 \times 10 = 2 \times 22 \times 10 = 440 \text{ cm}^2.$$

**Q27. (MODERN MATH - PERMUTATIONS & COMBINATIONS)** In how many ways can 5 boys and 3 girls be seated in a row such that no two girls sit together?

A) 14,400

B) 7,200

C) 2,880

D) 40,320

✓ **Correct Answer: Option A**

Seat 5 boys first:  $5! = 120$  ways.

This creates 6 available gaps for the girls: \_B\_B\_B\_B\_B\_.

Select and arrange 3 girls in 6 gaps:  ${}^6P_3 = 6 \times 5 \times 4 = 120$  ways.

Total ways =  $120 \times 120 = 14,400$ .

**Q28. (MODERN MATH - PROBABILITY)** Two unbiased dice are thrown. What is the probability that the sum of the numbers is a prime number?

A) 5/12

B) 15/36

C) 7/18

D) 1/2

✓ **Correct Answer: Option A**

Total outcomes = 36.

Possible prime sums from 2 dice: 2, 3, 5, 7, 11.

- Sum = 2: (1,1) [1]
- Sum = 3: (1,2), (2,1) [2]
- Sum = 5: (1,4), (2,3), (3,2), (4,1) [4]
- Sum = 7: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1) [6]
- Sum = 11: (5,6), (6,5) [2]

Total favorable =  $1 + 2 + 4 + 6 + 2 = 15$ .

Probability =  $15 / 36 = 5 / 12$ .

**Q29. (NUMBER SYSTEMS - UNIT DIGIT)** Find the unit digit of the expression  $7^{105}$ .

A) 7

B) 9

C) 3

D) 1

✓ **Correct Answer: Option A**

Cyclicity of 7 is 4 ( $7^1=7$ ,  $7^2=9$ ,  $7^3=3$ ,  $7^4=1$ ).

$105 \bmod 4 = 1$ .

Therefore, the unit digit is  $7^1 = 7$ .

| Company | 2023 | 2024 | 2025 |
|---------|------|------|------|
| Alpha   | 45   | 55   | 70   |
| Beta    | 60   | 75   | 85   |
| Gamma   | 50   | 65   | 90   |
| Delta   | 40   | 50   | 65   |

**Q30.** (DATA INTERPRETATION - TABLES) What is the percentage increase in sales for Company Gamma from 2023 to 2025?

- A) 40%
- B) 80%
- C) 60%
- D) 75%

✓ **Correct Answer: Option B**  
Gamma sales: 2023 = 50k, 2025 = 90k.  
Increase = 90 - 50 = 40k.  
Percentage Increase =  $(40 / 50) \times 100 = 80\%$ .

**DI SET: SMARTPHONE SALES ANALYSIS (Q30 - Q32)**  
The table shows the number of smartphones sold (in thousands) by four companies over three years.

| Company | 2023 | 2024 | 2025 |
|---------|------|------|------|
| Alpha   | 45   | 55   | 70   |
| Beta    | 60   | 75   | 85   |
| Gamma   | 50   | 65   | 90   |
| Delta   | 40   | 50   | 65   |

**Q31.** (DATA INTERPRETATION - AVERAGES) What is the average number of smartphones sold by all companies in 2024 (in thousands)?

- A) 61.25
- B) 60.5
- C) 62.5
- D) 65.0

✓ **Correct Answer: Option A**  
Total 2024 sales = 55 + 75 + 65 + 50 = 245k.  
Average =  $245 / 4 = 61.25$ k.

| Company | 2023 | 2024 | 2025 |
|---------|------|------|------|
| Alpha   | 45   | 55   | 70   |
| Beta    | 60   | 75   | 85   |
| Gamma   | 50   | 65   | 90   |
| Delta   | 40   | 50   | 65   |

**Q32.** (DATA INTERPRETATION - GROWTH COMPARISONS) Which company had the lowest percentage growth in sales from 2024 to 2025?

A) Alpha

B) Beta

C) Gamma

D) Delta

✓ **Correct Answer: Option B**

Growth from 2024 to 2025:

- Alpha:  $(15 / 55) = 27.27\%$
- Beta:  $(10 / 75) = 13.33\%$
- Gamma:  $(25 / 65) = 38.46\%$
- Delta:  $(15 / 50) = 30.00\%$

Beta had the lowest percentage growth (13.33%).

**Q33.** (DATA SUFFICIENCY - NUMBERS) Data Sufficiency: Is the integer  $x$  positive? Statement I:  $x^2 = 25$  Statement II:  $x^3 = 125$

A) Statement I alone is sufficient.

B) Statement II alone is sufficient.

C) Both statements together are necessary.

D) Neither statement is sufficient.

✓ **Correct Answer: Option B**

- Statement I:  $x = +5$  or  $-5$  (insufficient).
  - Statement II:  $x^3 = 125 \Rightarrow x = +5$  (definitely positive, unique answer).
- Hence, Statement II alone is sufficient.

**Q34.** (DATA SUFFICIENCY - GEOMETRY) Data Sufficiency: What is the area of a rectangle? Statement I: Its perimeter is 40 cm. Statement II: Its diagonal is 14.14 cm (approx  $10\sqrt{2}$ ).

A) Statement I alone is sufficient.

B) Statement II alone is sufficient.

C) Both statements together are necessary.

D) Neither statement is sufficient.

✓ **Correct Answer: Option C**

- St I:  $2(L + W) = 40 \Rightarrow L + W = 20$  (insufficient alone).
  - St II:  $L^2 + W^2 = (10\sqrt{2})^2 = 200$  (insufficient alone).
  - Combining:  $(L + W)^2 = L^2 + W^2 + 2LW \Rightarrow 400 = 200 + 2(\text{Area}) \Rightarrow \text{Area} = 100 \text{ cm}^2$ .
- Both statements together are necessary.



**Q35.** (DATA SUFFICIENCY - DIGIT LOGIC) **Data Sufficiency:** What is the two-digit number? **Statement I:** The sum of the digits is 8. **Statement II:** The difference between the digits is 2.

**A)** Statement I alone is sufficient.

**B)** Statement II alone is sufficient.

**C)** Both statements together are necessary.

**D)** Neither statement is sufficient.

✓ **Correct Answer: Option D**

- St I:  $x + y = 8$ .
  - St II:  $|x - y| = 2 \Rightarrow (x=5, y=3)$  gives 53, or  $(x=3, y=5)$  gives 35.
- Since two unique numbers are possible (35 and 53), neither statement (even combined) is sufficient.